

Introduction

The primary objective of this activity was to gain hands-on experience with **Power BI**, a robust business intelligence tool, to transform raw data into actionable visual insights. As part of our coursework at the **University of Science and Technology of Southern Philippines (USTP)**, this mini-workshop challenged us to move beyond familiar web-based platforms and explore the capabilities of desktop-integrated data analytics. The activity focused on loading group-specific datasets and generating unique visualizations to represent inventory or operational metrics effectively.

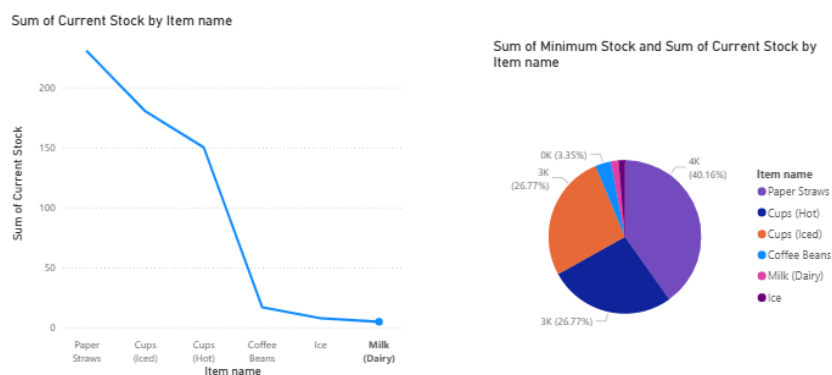
Methodology

The workshop followed a structured technical workflow to ensure data integrity and visual clarity:

- **Requirement Analysis:** We adhered to specific guidelines requiring at least two distinct visualizations, each utilizing a unique combination of data features to avoid redundancy.
- **Data Acquisition:** The process began by loading a group-specific **CSV file** into the Power BI Desktop environment.
- **Tool Selection:** Power BI was utilized as the primary integrated tool for this implementation, chosen for its powerful offline processing capabilities compared to web-based alternatives.
- **Visualization Construction:** Two primary charts were developed—a line graph and a pie chart—to represent different facets of the "Current Stock" and "Minimum Stock" levels across various inventory items.

Visual Documentation

The following image document the workshop requirements and the final output generated during the implementation phase:



This image outlines the workshop instructions, including the requirement to load our group dataset and produce unique visualizations.

Problem Solving

Transitioning to a new software ecosystem presented specific technical and cognitive hurdles that required quick adjustments:

- **The Challenge: UI Unfamiliarity:** Having previously used **Looker Studio** for academic projects, the Power BI User Interface (UI) felt unfamiliar and required a learning curve to locate specific formatting and data-binding tools.
- **The Solution:** We leveraged the intuitive drag-and-drop nature of the Power BI desktop app to explore the ribbon and visualization panes. By cross-referencing Looker Studio concepts with Power BI's "Fields" and "Format" tabs, we successfully mapped our existing knowledge to the new environment.
- **The Challenge: Access Requirements:** Power BI requires a specific account for full functionality, which can sometimes be a barrier.
- **The Solution:** Since we are students at **USTP**, we utilized our institutional credentials to bypass access issues, allowing for a seamless login and implementation process.

Conclusion

This workshop provided valuable insights into the practical advantages of desktop-based BI tools. A key learning was the **convenience of offline functionality**; unlike web apps, Power BI allowed us to build and refine our reports without a constant internet connection.

The experience highlighted the importance of adaptability in data science, demonstrating that while UIs may differ, the core principles of data visualization remain constant. We successfully delivered a concise dashboard that clearly differentiates stock levels for items like "Paper Straws" and "Coffee Beans," proving that with the right institutional support and tools, complex data can be made accessible and visually compelling.